

Solvency Field Theory V13.4-V2

Relational Interactions: Persistent Particle Roots and Complete Joint Configurations

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This replacement edition asks a narrower question than “derive all four forces” or “derive the Standard Model.” If SFT’s particles are persistent physical roots, can force, scattering, and reaction be obtained by solving the complete configuration made when those roots interact rather than by adding a separate force substance? The answer is yes at the architecture level and, in the strongest completed particle-level test, yes for the leading long-range electromagnetic interaction of two unchanged frozen Electron V1 roots. One constrained joint action generates the nonlinear configuration, its branchwise on-shell reduction gives the exact participant dynamics, the Schur/KKT response is the tangent of that nonlinear solve, and retarded elimination produces causal memory and open-channel response. The same force audit localizes rather than hides the remaining couplings and selectors: five continuous interaction/electroweak/Higgs-like debts remain explicit, while statistics, finite-overlap electron dynamics, full hadron roots, weak-chirality selection, generation masses, and detector actualization are handed to named child programs.

Replacement status. This V2 edition is the controlling standalone public narrative for the SFT V13.4 interaction sector. The earlier *SFT V13.4 Forces Full Derivation* remains a historical technical source. The controlling modern derivation is the *SFT V13.4 Relational Interaction Update Blend v0.11*, whose SHA-256 is listed in the source note.

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1 The narrow question

A useful way to expose the central SFT claim is to begin with a question ordinary notation can hide: when two physical objects interact, what is the thing that actually changes?

The usual language says that object A exerts a force on object B . That language is operationally effective, but it can encourage an unnecessary ontology: two independently complete objects plus a third entity called “the force” that travels between them. SFT uses a different starting point. A persistent particle is a lawful recurrent physical structure. When another particle, field support, apparatus, or other participant is present, the complete lawful configuration is no longer generally the product of the isolated solutions.

If A and B are individually admissible persistent roots, their interacting domain is generically

$$\Omega_{AB}^{\text{int}} \neq \Gamma_A \times \Gamma_B.$$

The interaction is therefore not an extra object added to A and B . It is the physical difference between the lawful joint continuation and the isolated continuations.

Force is the participant-side dynamical readout of a physical interaction: the change in a system’s lawful continuation produced by the complete joint configuration in which it participates, relative to its isolated continuation.

The same architecture classifies several familiar processes:

force = continuous identity-preserving interaction readout,
scattering = finite identity-preserving joint transition,
reaction/decay = identity-changing joint transition,
measurement = system–apparatus joint interaction producing distinguishable descendants.

The controlling lesson is simple:

Every time we stop treating “force” as a thing and instead solve the complete physical configuration, the equations get cleaner.

This is also why a universal definition such as $F^\mu = Dp^\mu/d\tau$ is too narrow to carry the ontology by itself. In gravitational free fall, for example, proper force may vanish even though the worldline differs from its flat-spacetime continuation. Force is one readout of interaction geometry, not the fundamental category.

This paper also keeps one hard ontology correction. There is no physical cosmic ledger, memory surface, or universal microscopic write operator. Older “write” language is retained only as bookkeeping shorthand for an irreversible realized segment of four-dimensional history. Real physical support propagates causally and geometry responds; the universe has nothing to write *on*.

2 The picture in plain language

Before the machinery, here is the whole idea in ordinary words.

Push two electrons toward each other and they push back. What is actually happening? The familiar story says that electron A exerts a force on electron B . That language is useful and the calculations built from it work extremely well. But it can make the force sound like a third physical thing added to the two particles. SFT asks whether that extra noun is necessary. Its answer is narrower: the interaction does not require a separate force substance over and above the particles, fields, geometry, and other physical response already present in the complete situation.

An isolated electron is one physical problem. In SFT Electron V1 it is a persistent recurrent structure with its own lawful support, current, charge/framing, spin response, and electromagnetic dressing. Put a second electron nearby and the correct problem is no longer “electron A alone” plus “electron B alone” with a correction mailed between them. The physical problem is the **joint configuration** made by both roots and the response they share. Changes still propagate causally through real physical support and fields; SFT does not introduce instantaneous action at a distance or a preferred universal present. What we call the force on either electron is the difference between that electron’s lawful continuation in the joint solution and the continuation it would have had in isolation.

For a static or conservative separated pair, there is an even plainer way to say this. The pair has a **joint energy bill**, and that bill depends on how far apart the electrons are. For like charges the interaction energy rises as the separation shrinks. The force is the slope of that shared energy with respect to separation. No extra force substance has to be added to the ontology: the bill belongs to the pair because the physical field configuration belongs to the pair.

That becomes more than a philosophical preference only if the actual SFT particle can survive the test. Electron V1 was frozen before this force calculation, with a checksum and a standing no-retuning rule. The interaction program then used two unchanged Electron V1 parents at long-range nonoverlap scope. It did not adjust the electron width, invent a new magnetic coefficient, add a velocity coupling, or fit a spin-orbit constant. The frozen charge and spin/current readouts were placed into one joint Maxwell response with one already-declared electromagnetic interaction normalization. The result was like-charge repulsion with the inverse-square force law, the spin magnetic interaction, the leading motion/retardation correction, and the spin-orbit structure including the Thomas/Wigner half correction. One interaction normalization was used throughout, with no new pair-specific continuous dial. In that limited but important sense, the particle paper and the force paper took each other's final exam and both passed.

The larger force program is organized the same way: **one architecture with different constitutive rooms**, not four identical mechanisms wearing different labels. Electromagnetism is presently the strongest completed room at leading long-range particle level. In the strong-sector candidate architecture, complete composites can be lawful even when fractional constituent projections are not independently complete asymptotic objects; that is a topology/admissibility result, not yet a derivation of confinement energy or string tension. The weak interaction is treated primarily as a branch-changing transition rather than a continuous push, while the physical origin of its chiral access rule remains open. Gravity is especially natural in this language because free fall can change a trajectory while producing zero ordinary proper force; the full particle-level gravitational continuation is nevertheless not claimed complete here.

The unfinished parts are printed in the ledger rather than hidden in the gears. At V13.4 force/electroweak/Higgs-like scope, five continuous quantities remain supplied rather than derived: the electromagnetic normalization, the strong normalization, one electroweak orientation ratio, one electroweak scale, and one Higgs-like radial stiffness. The frozen electron also retains its own parent-model support input and downstream unit anchor; those are not new force parameters, and the shared electromagnetic normalization is counted only once. The native explanation of identical-electron statistics remains a separate P6-L composition problem. And the full close-quarters two-electron solve—where the finite charge and magnetization structures genuinely overlap, radiate, scatter, and may probe compositeness—is the next mountain, not one this paper claims to have climbed.

Everything after this section is the mathematics of one instruction:

When things interact, stop looking for a separate force substance and solve what they have become together.

3 One persistent thing and two persistent things

The current SFT particle ontology distinguishes intrinsic structure from relational structure. A single closed persistent process can carry one-object invariants and readouts: rest-frame cadence, invariant mass readout, charge/framing class, spin magnitude/root class, and other properties of the maintained closure. Quantities such as separation, relative velocity, relative orientation, interaction energy, force, analyzer component, color-neutrality status, and detector outcome channel require a relation between at least two physical participants.

This paper therefore uses the rule

$$\boxed{\text{persistent root content} \neq \text{relational interaction content.}}$$

The distinction does not mean the two are disconnected. The purpose of V13.4-V2 is to show how an actual persistent root becomes an actual participant of a joint interaction without changing theories halfway through. The cleanest test is the electron.

The Level-1/Level-2 distinction used here is inherited from the current SFT particle ontology and V13.11 *The Cones and the Ruler* [10]. The relational architecture is also consistent with the V13.9-V3 conclusion that some physically meaningful facts belong irreducibly to relations rather than to either participant separately [9].

4 The frozen electron meets the force machine

SFT V13.12 freezes a Parameterized Stable Maintained Electron V1 [11]. At its stated scope, Electron V1 is one finite recurrent charged spin- $\frac{1}{2}$ structure whose finite support, source/current, electromagnetic dressing, spin/framing branch, local metric response, recurrence, and clock readout are treated as different faces of one maintained state.

The frozen electron supplies concrete participant data:

- a finite recurrent maintained support;
- a normalized physical source/current;
- a signed charge/framing branch $n_g = \pm 1$;
- a Pauli/Gordon rest branch with $S = \hbar/2$ and tree-level $g = 2$;
- stationary finite-domain nonradiating source behavior;
- nontrivial local closure-gravity response;
- one-count source bookkeeping;
- the intrinsic maintained-root parameter $\Gamma_\star$;
- one external electromagnetic normalization used by the charged dressing.

The electron freeze did not claim that two-particle interactions were already solved. Instead it imposed an interaction firewall: future interactions must be solved as joint states; they must not be modeled as an external force applied to a permanently isolated electron.

Let two isolated Electron V1 reference roots be $X_A^\star$ and $X_B^\star$. The physical interacting state is not

$$X_A^\star + X_B^\star + \text{an external force.}$$

It is one complete joint solution,

$$X_{AB}^\star(R) = \text{Solve}[\delta S_J[X_A, X_B, Y] = 0],$$

with isolation boundary

$$X_{AB}^\star(R) \longrightarrow X_A^\star \oplus X_B^\star \quad (R \rightarrow \infty).$$

At finite separation the complete support/field/geometry configuration generally differs from the two isolated solutions. That does not mean either electron has become a different particle. The same persistent roots now participate in a different relational configuration. This is the Root–Surplus distinction:

$$\boxed{\text{persistent root invariant content} \neq \text{surplus relational/kinematic content.}}$$

As long as the process remains branch-preserving, additional interaction energy may appear in relative four-momentum, joint fields, radiation, or other relational content without continuously changing the electron's invariant root identity.

5 One total electromagnetic response

A potential double-counting problem appears immediately because the isolated Electron V1 is already electromagnetically dressed. The wrong construction would be

$$(\text{electron A} + \text{self-field}) + (\text{electron B} + \text{self-field}) + (\text{new interaction field}).$$

That would treat the interaction field as a second copy of a response sector already present in the charged particle description.

The correct construction uses one total Maxwell response sourced by the total physical current,

$$\boxed{J^\mu = J_A^\mu + J_B^\mu.}$$

The isolated self-dressings are reference solutions of the same field sector. They are not additional fields that coexist with a third force field.

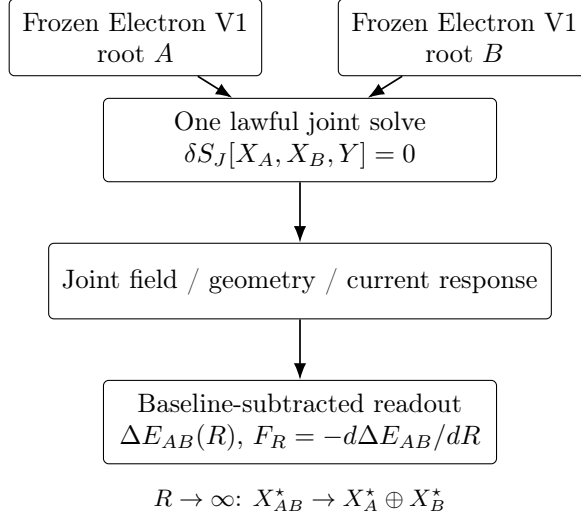


Figure 1: The flagship sewing rule. The interacting pair is one joint configuration with the isolated Electron V1 roots as its large-separation boundary, not two pointwise frozen self-field solutions plus a third force field.

At leading linear Maxwell scope, write the on-shell field response schematically as

$$S_{\text{EM}}^*[J] \propto \frac{1}{2} \langle J, G_{\text{EM}} J \rangle.$$

Then

$$\begin{aligned} \Delta S_{AB}^{\text{EM}} &= S_{\text{EM}}^*[J_A + J_B] - S_{\text{EM}}^*[J_A] - S_{\text{EM}}^*[J_B] \\ &\propto \langle J_A, G_{\text{EM}} J_B \rangle. \end{aligned}$$

The interaction is therefore the cross-response of the same total field action whose self-response participates in the isolated charged roots.

For a conservative separated branch, define

$$\Delta E_{AB}(R) = E[X_{AB}^*(R)] - E[X_A^*] - E[X_B^*],$$

so that

$$F_R = -\frac{d\Delta E_{AB}}{dR}.$$

This is the particle-level realization of the relational force definition.

There is one additional consistency requirement. The electromagnetic normalization used in the charged Electron V1 dressing and in the pair interaction is one common interface coefficient at the same declared reference scale and convention:

$$\alpha_{\text{Electron dressing}}(\mu_0) = \alpha_{\text{pair interaction}}(\mu_0) \equiv \alpha(\mu_0).$$

If future work derives or varies α , the charged electron must be re-solved as a lawful family $X_e^*(\alpha)$ rather than holding the exact frozen dressing fixed while changing the global Maxwell stiffness externally.

6 The first real integration result

The V13.4 program performed the first direct particle/interaction regression with two unchanged frozen Electron V1 parents. The stop rule was strict: no interaction result could retune the frozen electron width; no new magnetic coefficient, velocity coupling, or spin-orbit coefficient could be inserted; and no observed two-electron target could be imported as a parent result.

At long-range nonoverlap scope the pair problem uses the single already-declared electromagnetic coefficient

$$C_{\text{EM}} = \frac{q_0^2}{4\pi K_{\text{EM}}}.$$

In conventional normalization $C_{\text{EM}} = \alpha \hbar c$, but numerical α remains one declared reference-scale interaction input.

Static electric monopole

With $J^\mu = J_A^\mu + J_B^\mu$, the joint Maxwell action contains the cross term between the two normalized sources. Outside the finite supports,

$$U_{AB}(R) = \frac{Q_A Q_B}{4\pi K_{\text{EM}} R} + O(R^{-2}).$$

For two electrons, $Q_A Q_B = q_0^2 > 0$, so

$$U_{ee}(R) = \frac{C_{\text{EM}}}{R}, \quad \mathbf{F}_{ee}(R) = \frac{C_{\text{EM}}}{R^2} \hat{\mathbf{R}}.$$

The force is repulsive. No new two-electron coefficient appears.

Magnetic dipole interaction

The frozen Pauli/Gordon branch supplies

$$\boldsymbol{\mu} = \frac{q}{m} \mathbf{S}, \quad S = \hbar/2, \quad g_{\text{tree}} = 2,$$

with the conventional Dirac/Pauli $g = 2$ correspondence noted in Ref. [1]. The far-field pair interaction is

$$U_{ss} = \frac{C_{\text{EM}}}{m^2 c^2 R^3} \left[\mathbf{S}_A \cdot \mathbf{S}_B - 3(\mathbf{S}_A \cdot \hat{\mathbf{R}})(\mathbf{S}_B \cdot \hat{\mathbf{R}}) \right].$$

The same EM normalization controls the Coulomb and magnetic terms.

Motion, retardation, and spin-orbit structure

Expanding the same retarded joint Maxwell configuration through conservative order v^2/c^2 gives the Darwin class [2],

$$L_{AB} = -\frac{C_{\text{EM}}}{R} + \frac{C_{\text{EM}}}{2c^2 R} \left[\mathbf{v}_A \cdot \mathbf{v}_B + (\mathbf{v}_A \cdot \hat{\mathbf{R}})(\mathbf{v}_B \cdot \hat{\mathbf{R}}) \right].$$

Magnetic current-current response and longitudinal retardation are one covariant correction; no additional velocity coupling is required.

Using the same frozen $g = 2$ Pauli moment, Lorentz kinematics, and the Thomas/Wigner frame correction [3] gives, in the equal-mass center-of-momentum convention used in the audit,

$$H_{SO}^{\text{COM}} = -\frac{3C_{\text{EM}}}{2m^2 c^2 R^3} \mathbf{L} \cdot (\mathbf{S}_A + \mathbf{S}_B).$$

The broader conventional two-electron retardation/spin operator context is associated with the Breit interaction [4]. V13.4-V2 does not claim a complete Breit-Pauli or QED derivation.

The safe integrated statement is therefore:

Two unchanged frozen Electron V1 roots reproduce the leading long-range Coulomb, convective-retardation, spin-spin, and spin-orbit interaction classes with one common electromagnetic normalization and no parent-electron retuning.

This regression uses the frozen parent's integrated source normalization, charge/framing readout, Pauli magnetic moment, kinematics, and common EM normalization. It is not yet a full nonlinear finite-overlap solve of every detailed Electron V1 support coordinate.

7 One constrained joint-action machine

The electron integration is one realization of a more general architecture. Let the complete physical configuration be X . SFT uses one constrained joint action

$$S_J[X] = \int_{\mathcal{M}} \mathcal{L}_{\text{native}}(X, \nabla X, g, \dots) d^4x,$$

with lawful domain

$$\mathcal{D}_J = \{X : \mathcal{C}_a[X] = 0, \mathcal{I}_b[X] \geq 0\}.$$

Physical evolution satisfies

$$\boxed{\delta S_J[X] = 0, \quad X \in \mathcal{D}_J.}$$

The transition map is not a new ontic machine. It is simply the induced nonlinear solution map

$$\boxed{\mathcal{T}_J : X_{\text{in}} \mapsto X_{\text{out}}.}$$

Split the configuration locally into retained participant variables x and responding variables y . On a smooth lawful constitutive branch,

$$\frac{\delta S_J[x, y]}{\delta y} = 0 \quad \Rightarrow \quad y = y_\star[x].$$

Define the branchwise on-shell action

$$\boxed{S_{\text{eff}}[x] = S_J[x, y_\star[x]].}$$

Then the response-stationarity term drops out of the variation,

$$\boxed{\frac{\delta S_{\text{eff}}}{\delta x} = \left. \frac{\delta S_J}{\delta x} \right|_{y_\star} .}$$

Solving the responding configuration first and then reading the participant dynamics is therefore exactly equivalent, branchwise, to varying the solved joint configuration.

The theorem is not a branch selector. If several lawful solutions $y_\star^{(a)}[x]$ exist, several branchwise reduced actions exist. Formation, bifurcation, transition completion, history/context, or admissibility must determine which branch is physically realized.

8 The distributor is the tangent of the engine

Differentiate the solved response condition:

$$H_{yx} + H_{yy} D y_\star = 0.$$

If the response block is invertible on the lawful physical quotient,

$$\boxed{D y_\star = -H_{yy}^{-1} H_{yx} .}$$

The second variation of the on-shell participant action is

$$\boxed{H_{\text{eff}} = H_{xx} - H_{xy} H_{yy}^{-1} H_{yx} .}$$

Thus the Schur response is not a separate linear ansatz. It is the tangent limit of the exact nonlinear joint solve. For constrained systems, the same statement is written on the lawful constraint manifold or with the KKT-augmented Hessian so that constraint reaction is preserved.

For causal dynamics, the static inverse is replaced by the retarded physical inverse,

$$\boxed{\Delta \mathcal{J}_x^R = -\mathcal{J}_{xy} (\mathcal{J}_{yy}^R)^{-1} \mathcal{J}_{yx} .}$$

The nonlinear engine, the linear distributor, and the retarded response are one machine viewed at different resolutions.

At a solved configuration the action-owned cross-blocks obey, after gauge and constraint reduction,

$$H_{yx} = H_{xy}^\dagger.$$

The participant-to-response coupling and the response-to-participant backreaction are therefore adjoints of the same physical coupling. Separate outbound and backreaction normalizations are not independent knobs if both are claimed to descend from one joint action.

If H_{yy} develops a zero mode, the Schur inverse fails locally. This is a diagnostic of a bifurcation, instability, zero mode, or opening/closing channel; the full nonlinear solve must then be used through the singular point. It is not automatically a failure of the joint architecture.

9 The constitutive plug

The common-action architecture does not imply that all sectors have the same microscopic physics. Define the constitutive plug of sector/process s as

$$\mathfrak{C}_s = (\mathcal{Y}_s, \mathcal{D}_s, \mathcal{K}_s, \lambda_s).$$

Here $\mathcal{Y}_s$ is the actual responding physical support/variables, $\mathcal{D}_s$ is the lawful topology/representation/admissibility domain, $\mathcal{K}_s$ is the sector-specific nonlinear constitutive response and cross-coupling inside the complete joint action, and λ_s contains only genuinely unresolved declared normalization/scale data.

The controlling distinction is

$$\text{same constitutive socket geometry} \neq \text{same microscopic constitutive plug.}$$

Electromagnetism, strong response, weak response, Higgs-like collective response, flavor response, and detector completion may therefore be physically different while entering one general joint-action architecture.

It is useful to classify the remaining debts as

$$\begin{aligned} \mathbf{D}\text{-debt} &= \text{domain / representation / composition selector,} \\ \mathbf{K}\text{-debt} &= \text{nonlinear constitutive kernel / root / cross-response solve,} \\ \mathbf{N}\text{-debt} &= \text{normalization / scale.} \end{aligned}$$

10 Electromagnetism: the strongest completed room

Let A_μ be the local connection associated with the compact charge/framing response and

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Under local $U(1)$ relabeling, Lorentz invariance, smooth weak-field response, stability, parity-even bulk dynamics, and leading derivative/power order, the unique leading parity-even dynamical bulk term is

$$S_{\text{EM}} = -\frac{K_{\text{EM}}}{4} \int F_{\mu\nu} F^{\mu\nu} d^4x.$$

Stability fixes $K_{\text{EM}} > 0$ but not its magnitude. Canonical normalization gives the physical low-energy strength as the invariant ratio

$$\alpha \sim \frac{q_\theta^2}{4\pi K_{\text{EM}}}$$

up to convention.

The audit closes several tempting selectors. Compact charge topology alone, local $U(1)$ symmetry alone, isolated Electron V1 maintenance, internal-to-exterior connection continuity, a fixed source plus its Maxwell exterior, running without an independently fixed boundary condition, and fixed-point expressions whose coefficient ratio is itself underived do not select the numerical value of α .

The EM sector therefore contains one reference-scale normalization,

$$\boxed{\alpha(\mu_0),}$$

with public status PASS WITH ONE FREE PARAMETER. The failure of isolated electron maintenance to select α is therefore diagnostic of the coupling belonging to the coupled particle–field interaction sector rather than a failed electron-maintenance selector.

11 Charge, fractional structure, and complete composites

SFT separates mass cadence from charge/framing structure. A complete isolated compact $U(1)$ charge/framing map has integer degree,

$$\boxed{\frac{1}{2\pi} \oint \mathfrak{h} \in \mathbb{Z}.}$$

At leading formation order, the winding selector is

$$\boxed{n_\star = \arg \min_{n \in \Lambda_{\text{lawful}}} S_{\text{form}}(n),}$$

conditional on the physical boundary/composite class.

For a standalone doublet class the least lawful winding gives $|n_Y| = 3$, corresponding in one orientation to $(0, -1)$. For a primitive fractional doublet class the least lawful odd winding gives $|n_Y| = 1$, corresponding in one orientation to $(2/3, -1/3)$. A complete standalone $1/3$ or $2/3$ winding is not independently allowed as an isolated compact charge closure.

If a fractional residue sector exists, Z_2 is too small to distinguish nonzero conjugate orientations because $1 = -1 \pmod{2}$. The smallest oriented cyclic quotient is therefore Z_3 . Configurations such as $1 + 2 = 0 \pmod{3}$ and $1 + 1 + 1 = 0 \pmod{3}$ can be globally complete while their nonzero projections remain individually incomplete. Hence

$$\boxed{\mathcal{D}_{\text{fractional comp}} \not\subseteq \Gamma_1^{\text{complete}} \times \Gamma_2^{\text{complete}} \times \dots .}$$

The mature statement is that the complete composite may be a lawful physical object even though fractional constituent projections are not independently complete asymptotic particles. This is an admissibility result, not yet a binding-energy law, string tension, hadron spectrum, or complete QCD derivation.

12 Strong interaction architecture

A genuinely complex three-component internal response fiber with local basis freedom supports a trace-free unitary connection class. In QFT/SM this mathematical structure is the $SU(3)$ Yang–Mills connection introduced by Yang and Mills [5]; in SFT the native interpretation is a relational connection on a three-component port/color response space rather than an imported field ontology.

The leading curvature class may be written

$$\boxed{S_s = -\frac{1}{2g_s^2} \int \text{Tr}(\mathcal{F} \wedge *\mathcal{F}) + S_{\text{comp}}.}$$

The non-Abelian self-interaction is already inside $\mathcal{F}$. Separate cubic and quartic connection couplings are therefore not independent knobs.

The audit also rejects an independently inserted string-tension parameter, arbitrary scalar current multiplier, and separate port-attraction coefficient. One reference normalization remains,

$$\boxed{g_s(\mu_0),}$$

or an equivalent $\alpha_s(\mu_0) / \Lambda_s$ parametrization. Topology, source normalization, and gauge self-interaction do not by themselves select its value.

Public status: a CONDITIONAL strong operator class with ONE FREE REFERENCE NORMALIZATION. The stable anisotropic port-resolved hadron root and string-tension output remain a child nonlinear BVP.

13 Weak transition and electroweak response

The weak sector is naturally described as a local joint transition connecting distinct persistent closure sectors. Let Ξ be the eventual native Hermitian chirality-orientation involution,

$$\Xi^\dagger = \Xi, \quad \Xi^2 = I.$$

In QFT/SM the conventional mathematical role is occupied by γ_5 ; SFT does not import γ_5 as ontology.

The most general leading chirality-only access map is $\Pi = aI + b\Xi$. If charged-current access is a nontrivial proper projector, $\Pi^2 = \Pi$, then

$$\Pi_\pm = \frac{1}{2}(I \pm \Xi).$$

No continuous “amount of chirality” parameter survives. However, current SFT legality conditions do not force $\Pi_\pm \otimes T_\pm$ over the vectorlike alternative $I \otimes T_\pm$. If the complete action and admissible domain are parity symmetric, parity maps one chiral branch into an equally lawful mirror branch. The physical origin of chiral charged-current access therefore remains one structural transition-law debt.

Write the electroweak response Hessian as

$$\mathcal{M}_{\text{EW}}^2 = \mu_{\text{EW}}^2 \widehat{\mathcal{M}}_{\text{EW}}^2.$$

A common rescaling changes absolute masses while preserving rank, kernel, eigenvectors, mixing orientation, and mass ratios. The shape of the Hessian cannot determine its own absolute scale. Positive response curvature gives finite correlation length,

$$(-\nabla^2 + m_i^2)W_i = J_i, \quad \ell_i = \frac{\hbar}{m_i c}.$$

At V13.4 scope, μ_{EW} remains one free scale. Once a closed particle supplies a physical Compton ruler, the deeper target can be retyped as a dimensionless comparison $\xi_{\text{EW}|A} = \mu_{\text{EW}}/m_A$.

14 The minimal same-fiber branch

A single collective amplitude does not by itself force the W/Z relation. The stronger hypothesis is that the collective electroweak response lives in the already-owned irreducible rank-two complex weak branch fiber,

$$E \simeq \mathbb{C}^2.$$

For a positive Hermitian branch-response metric invariant under local $SU(2)$ -type basis covariance,

$$U^\dagger h U = h \quad \forall U \in SU(2),$$

irreducibility gives

$$h = K_L I.$$

There is therefore no independent leading quadratic stiffness assigned separately to T_1, T_2, T_3 .

With neutral orientation selected, $Q\Phi_\star = 0$, the same invariant response norm gives

$$m_W^2 \propto g^2, \quad m_Z^2 \propto g^2 + g'^2, \quad \tan \theta_W = \frac{g'}{g},$$

and hence

$$m_W = m_Z \cos \theta_W.$$

This result is **CONDITIONAL** on the minimal fundamental same-fiber response. SFT has not yet derived why nature realizes that representation rather than a more general anisotropic or higher-representation response.

Use the geometric-connection convention

$$D = d + i\mathcal{W}^a T_a + i\mathcal{B}Y.$$

Along the surviving electromagnetic diagonal, the leading quadratic response gives

$$K_{\text{EM}} = K_W + K_Y.$$

With $r_{EW} = K_W/K_Y$, canonical normalization yields

$$e = g_W \sin \theta_W = g_Y \cos \theta_W,$$

so

$$g_W = \frac{e}{\sin \theta_W}, \quad g_Y = \frac{e}{\cos \theta_W}.$$

Independent leading g_W and g_Y normalizations are therefore deleted as free parameters once the EM normalization and one EW relative orientation are supplied. This reproduces the familiar electroweak coupling geometry associated conventionally with Weinberg's model [7], but the SFT claim is a native response interpretation, not an import of Standard Model ontology.

15 The Higgs-like amplitude response

Under the minimal same-fiber hypothesis, $\Phi \in \mathbb{C}^2$. Local branch-basis covariance implies the derivative-free scalar self-response depends only on

$$\rho = \Phi^\dagger \Phi.$$

A nonzero realized orientation breaks three orientation directions while preserving one $U(1)_{EM}$ direction. Since $\mathbb{C}^2$ has four real components, exactly one gauge-invariant radial/amplitude direction remains.

In QFT/SM, a scalar amplitude associated with electroweak symmetry breaking is conventionally described by Higgs-type field language [6,7]. In SFT, the native claim is narrower: a collective same-fiber response possesses one radial amplitude mode. It is not asserted to be a fundamental ontic Higgs field.

Write

$$\Phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h \end{pmatrix}, \quad \rho_\star = \frac{v^2}{2}.$$

At equilibrium $V'(\rho_\star) = 0$, and radial curvature gives

$$m_H^2 = v^2 V''(\rho_\star).$$

But equilibrium does not determine radial curvature. Two lawful invariant potentials can share the same equilibrium and W/Z geometry while having different second derivatives. Define

$$\zeta_H = \frac{m_H^2}{m_W^2}.$$

Stability gives only $\zeta_H > 0$. Its magnitude remains one dimensionless self-response parameter. There is no separate Higgs-like absolute mass scale beyond the common EW scale, and the leading HWW/HZZ response normalizations are derivatives of the same W/Z loading term rather than independent leading knobs.

16 Three tangent modes, hierarchy, and flavor mixing

For one timelike persistent root in $3 + 1$ spacetime, the physical normal space has rank

$$\dim N_p = 4 - 1 = 3.$$

The physical tangent response Hessian therefore has exactly three tangent spectral modes counting multiplicity, with no fourth independent tangent eigenvector unless the physical state space is enlarged. A cyclic three-mode response gives

$$u_k(\beta) = 1 - \cos \left(\beta + \frac{2\pi k}{3} \right), \quad k = 0, 1, 2.$$

For generic β , the three tangent rigidities are distinct and positive.

Interpreting these three modes as the physical generation sectors is a downstream hypothesis, not part of the mode-count theorem. Nor may the tangent rigidity itself simply be identified with mass. A generation-blind

collective-response operator $I_{\text{gen}} \otimes H_{\text{EW/H}}$ loads all three modes equally and cannot produce a hierarchy. A genuine hierarchy therefore requires a nontrivial generation–collective-response cross-block,

$$Y_{gH} \not\propto I_3.$$

There is also a structural mixing constraint. If the mass-response and weak-transition family operators are functions only of the same cyclic incidence structure, they commute and share the same Fourier basis. Therefore **nontrivial CKM/PMNS-style mixing requires a second family-sensitive physical response that does not commute with the first**. This constrains where flavor mixing is allowed to originate; it does not predict the observed mixing angles or phases.

17 Retarded response, memory, and open channels

The joint-action machinery explains why delayed or memory-like effective stresses can appear without introducing a fundamental lag substance. Consider an enlarged self-adjoint system

$$i\hbar \frac{d}{dt} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} H_{PP} & V \\ V^\dagger & H_{QQ} \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix}.$$

Eliminating q gives an exact retarded memory kernel. In energy space,

$$\Sigma^R(E) = V(E - H_{QQ} + i0)^{-1} V^\dagger = \Delta(E) - \frac{i}{2} \Gamma(E),$$

with

$$\Gamma(E) = 2\pi V \delta(E - H_{QQ}) V^\dagger \succeq 0.$$

In a slow long-wavelength regime,

$$\Sigma^R(\omega, \mathbf{k}) = \Sigma_0 + c_k k^2 - i\eta\omega + c_\omega \omega^2 + \dots,$$

which generates static response, gradient response, damping, relaxation, and telegraph/viscoelastic effective behavior. The older SFT “lag” intuition therefore survives only as a coarse-grained effective readout of finite causal response, not a stored physical deficit.

For a channel response $r_i = P_i R_J^{\text{ret}} \eta_J$, define the positive response capacity

$$\mu_i = \langle r_i, G_i r_i \rangle = \|r_i\|_{G_i}^2.$$

At stationary retarded-response scope the open-channel current is

$$J_i^{P \rightarrow Q} = \frac{1}{\hbar} \langle p, \Gamma_i p \rangle.$$

Integrated positive completion effects then occupy the mathematical class ordinarily represented by POVM/effect language in quantum mechanics. In SFT the native interpretation is integrated positive completion current derived from causal open-channel dynamics. The remaining measurement debt is the full nonlinear identification between microscopic detector basins and reduced open sectors, together with objective single-event completion.

18 Statistics, finite overlap, and selected hostile controls

The long-range two-electron force stack does not require the electron exchange sign to derive Coulomb, magnetic, Darwin, or spin–orbit response. For identical electrons, exchange has two discrete representation classes $\eta_e = \pm 1$. Current SFT premises do not yet select the fermionic sign. Inside the bilinear composition class,

$$C(u, u) = 0 \quad \forall u \quad \Longleftrightarrow \quad C(u, v) = -C(v, u),$$

but the physical origin of that self-composition nullity remains localized to the P6-L local relativistic root-composition program. This is the native problem corresponding to the conventional relativistic spin–statistics theorem [8]. V13.4-V2 does not claim to have reproduced that theorem.

A mature form of the older SFT “gear” idea survives only as a finite-overlap relational candidate. Let $\Delta_g \sim \Delta_g + 2\pi$ be the relative compact framing angle. For a real exchange-even local overlap cost,

$$V(\Delta_g) = \sum_{n \geq 1} \kappa_n [1 - \cos(n\Delta_g)].$$

Under a minimal leading-harmonic rule,

$$V_{\text{gear}}^{(1)} = \kappa_g [1 - \cos \Delta_g], \quad \kappa_g > 0.$$

This is a candidate finite-overlap framing-compatibility term, not the origin of long-range Coulomb strength, not a selector for α , and not an isolated electron rest-mass spring.

Several hostile controls are now frozen because they prevent future overgrowth:

Route	Ruling
Two pointwise frozen self-field electrons plus a third force field	Rejected: double-counts one response sector.
Independent source and backreaction couplings inside one action	Rejected: the cross-blocks are adjoints of one physical coupling.
Topology alone selects α or g_s	Rejected: operator class does not fix response normalization.
Generation-blind same-fiber loading creates hierarchy	Rejected: identical loading cannot split generations.
Equilibrium geometry fixes Higgs-like radial curvature	Rejected: the same equilibrium admits continuously different second derivatives.
A singular Schur block automatically means the theory fails	Rejected: it can diagnose a bifurcation/zero mode requiring the full nonlinear solve.

19 Parameter ledger and child programs

Because this paper integrates an actual Electron V1 parent with the force architecture, two ledgers must be kept distinct.

Parent Electron V1 interface/debt ledger

Electron V1 enters V13.4 as an already-frozen parent model. Its relevant non-derived interfaces include the maintained-root formation/support input Γ_* , a downstream dimensional rest-scale anchor, and the common electromagnetic normalization $\alpha(\mu_0)$. The first two are not new force-sector parameters. The third is counted once globally, not once in Electron V1 and again in V13.4.

V13.4 interaction/electroweak/Higgs-like ledger

At leading scope the remaining continuous ledger is

$$\alpha(\mu_0), \quad g_s(\mu_0), \quad r_{\text{EW}}, \quad \mu_{\text{EW}}, \quad \zeta_H.$$

Quantity	Role	Status
$\alpha(\mu_0)$	Electromagnetic reference normalization	PASS WITH ONE FREE PARAMETER
$g_s(\mu_0)$	Strong reference normalization	PASS WITH ONE FREE PARAMETER
r_{EW}	Electroweak relative stiffness/orientation	PASS WITH ONE FREE PARAMETER
μ_{EW}	Common electroweak response scale	PASS WITH ONE FREE SCALE
ζ_H	Higgs-like radial self-curvature	PASS WITH ONE FREE DIMENSIONLESS PARAMETER

Several historically separate knobs are no longer independent at leading scope: g_W , g_Y , an independent m_W/m_Z ratio inside the minimal same-fiber branch, a separate Higgs-like absolute mass scale, leading HWW/HZZ normalizations, separate strong cubic/quartic couplings, a separate string-tension knob, separate source/backreaction couplings, and a separate two-electron interaction coefficient.

The named child programs are equally important to the claim boundary:

- **finite-source two-electron completion:** charge/magnetization overlap, contact regularization, high- q form factors, radiation reaction, relativistic scattering;
- **P6-L statistics:** local identical-root composition sign/self-composition nullity;
- **particle spectrum:** nonlinear stable generations, mass ratios, Y_{gH} , flavor mixing;
- **strong composite program:** stable hadron root, string tension, scale/running, spectrum;
- **weak-transition compatibility:** physical selector for chiral rather than vectorlike charged-current access;
- **same-fiber representation selector:** why nature realizes the minimal rank-two collective response;
- **absolute/relational scale:** derive μ_{EW}/m_{root} or a common SFT ruler;
- **Higgs-like constitutive response:** derive ζ_H if possible;
- **detector actualization:** nonlinear basin/open-sector equivalence and single-event completion;
- **precision correspondence:** amplitudes, widths, loops, running, anomalous moments, CKM/PMNS, high- q limits, and precision S-matrix comparison.

20 What this revision changed

V13.4-V2 is not merely a relabeling of the earlier force narrative. Five conceptual changes control the replacement.

1. **Force stopped being a substance.** The primary object is the complete joint configuration; force is a participant readout.
2. **The particle and force programs were sewn together.** Two unchanged Electron V1 roots already produce the leading long-range electromagnetic interaction stack with one shared normalization.
3. **The nonlinear and linear machines were identified.** Exact on-shell elimination gives participant dynamics; the Schur/KKT response is its tangent.
4. **Sector differences were moved into constitutive plugs.** The shared interface no longer implies a claim that EM, strong, weak, flavor, and detector physics share one microscopic mechanism.
5. **Parameters were localized instead of hidden.** Operator-class results are separated sharply from normalization, scale, representation, and branch-selector debts.

The revised architecture is consequently smaller. Several old force-specific coefficients disappear, several former “force” questions become representation or constitutive questions, and the remaining failures are easier to name.

21 Claim boundary and falsification

What V13.4-V2 claims

This revision claims:

- a relational force ontology based on complete joint configurations;
- one constrained joint-action solution architecture;
- exact branchwise nonlinear constitutive elimination;
- Schur/KKT response as the tangent of the nonlinear on-shell solve;
- causal retarded response and open-channel current within the same machine;
- no independent source/backreaction coupling normalization for one action-owned interaction;
- a constitutive plug contract for sector-specific physics;
- leading Maxwell bulk operator closure with one EM normalization;
- conditional charge winding and Z_3 composite architecture;
- a conditional strong $SU(3)$ -type operator class with one strong normalization;

- a weak chiral projector class with one structural selector debt;
- a conditional minimal same-fiber electroweak branch with $m_W = m_Z \cos \theta_W$;
- deletion of independent leading g_W, g_Y normalizations;
- one Higgs-like radial amplitude mode in the minimal branch with one radial self-curvature parameter;
- exactly three tangent normal-response modes;
- exact retarded memory from causal reduction;
- a particle/force integration pass using two unchanged frozen Electron V1 roots at leading long-range EM scope.

What V13.4-V2 does not claim

This revision does not claim a complete derivation of the Standard Model; numerical α , α_s , θ_W , the absolute weak scale, or the Higgs-like mass; a native spin–statistics theorem; an exact three-generation massive spectrum; completed QCD confinement/string tension; a derived weak-chirality selector; a completed detector actualization theory; a full QED/Breit–Pauli/S-matrix derivation; a high- q pointlikeness/compositeness theorem for Electron V1; a whole-space two-electron gravity theorem; or one universal microscopic constitutive action already solved for every sector.

The architecture is falsifiable. It would be undermined if the full finite-source two-electron solve cannot connect continuously to the frozen Electron V1 parents without retuning intrinsic electron parameters; if correct two-electron physics requires a second independent EM coupling beyond the normalization already used by Electron V1; if action-owned source and backreaction require independent couplings; if no lawful replacement survives failure of the minimal same-fiber electroweak branch; if the strong composite topology cannot support a stable full root even with the allowed constitutive completion; or if the theory fails precision correspondence once amplitudes, widths, loops, running, and high- q behavior are actually calculated.

Conversely, future successes must not be obtained by silently adding sector-specific fitted constants or by changing the frozen parent particle only to hit an interaction target.

22 The sentence

SFT V13.4-V2 treats force, scattering, and reaction as readouts of complete constrained joint physical configurations rather than as independent substances. Persistent particle roots supply the participants; one joint action supplies the nonlinear combined configuration; branchwise on-shell elimination gives the exact participant dynamics; and retarded reduction gives causal memory and open-channel response. The strongest completed particle-level integration uses two unchanged frozen Electron V1 roots and reproduces the leading long-range Coulomb, magnetic, Darwin/retardation, and spin–orbit interaction classes with one common electromagnetic normalization. Charge, strong, weak, electroweak, Higgs-like, flavor, statistics, detector, gravity, and precision extensions remain at explicitly different levels of completion, with their surviving parameters and structural debts printed in the ledger rather than hidden in the machinery.

Acknowledgments

The present V13.4-V2 replacement paper is the public narrative of the relational force reaudit developed by Shaun Fosmark and GPT-5.6 SOL. The earlier V13.4 force derivation, the V13.12 Electron V1 freeze, the two-electron flagship gates, the electroweak/Higgs-like parameter audits, the constitutive-interface gates, and the final Electron/Force Integration gate are absorbed here at their final claim boundaries rather than reproduced in discovery order.

Claude/Opus (Fable) provided independent hostile review, proposed the vivid plain-language framing incorporated into Sec. 2 after final claim-boundary audit, and identified the final public corrections concerning the Maxwell uniqueness assumptions and the noncommuting flavor-mixing corollary. Earlier SFT collaborators and archived gate programs remain credited in their source documents; this replacement preserves those contributions without requiring the reader to reconstruct the historical sequence.

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Controlling derivation: SFT V13.4 Relational Interaction Update Blend v0.11 — Electron/Force Integration

Derivation SHA-256: cd561c041eab7c067a46a1d6c56f3670700aa862685be0d67cf81732cec1100e

Public narrative status: V13.4-V2 replacement edition v1.0; earlier V13.4 force narrative retained as historical record.